# The Useful Emptiness: An Inboard Flap Gap as an Energy Concentrator for a Distributed-Propulsion eSTOL

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**A 11.07-m-span, 2,600-lb ultra-STOL (uSTOL) aircraft with a 10-propeller distributed-electric propulsion array is studied across cruise, takeoff, and landing using high-fidelity RANS CFD (Flow360, actuator-disk thrust model). Starting from the naive low-htail / continuous-flap baseline (the X-57-class layout, known to have stability shortcomings at high blowing), we walk through a four-step configurational *evolution*. Step 1 raises the stabiliser to a high T-tail – adequate at today's published uSTOL approach envelopes but, when pushed to the slower approach speeds and more aggressive flap settings of this study, only weakly stable in takeoff and mildly unstable in landing. Subsequent steps then: (i) an inboard flap gap that drives a centerline upwash column from the streamwise vortices shed at the flap discontinuities; (ii) the return of the horizontal tail into the prop slipstream column for a 4–9× local-q boost; and (iii) the shortening of the tail boom – enabled by the resulting stability headroom – from $4\,c$ to $2\,c$ aft of the wing trailing edge. The final v3 configuration is stable in all phases (measured SM = +15/+19/+21% MAC cruise/takeoff/landing on the partial v3 sweep so far), trims the entire landing envelope including high-thrust go-around within the htail stall margin, allows aggressive landing rotation without tail strike, and reduces both landing roll and go-around pitch transient. The full paper will add nonlinear phugoid dynamic simulations for the three principal configurations and a CFD-resolved ground-effect landing maneuver.**

## Nomenclature

| | |
|---|---|
| $\alpha$ | body angle of attack |
| $\theta_{ht}$ | horizontal-tail rotation (positive: TE down) |
| $T/L$ | delivered total thrust normalised by total lift; $L$ includes the thrust's vertical component, so $T/L = C_T/C_{L,\text{total}}$ in the wind frame; pure CFD-derived dimensionless ratio |
| $C_L, C_D, C_{m_{\text{CG}}}$ | lift, drag, and pitching-moment coefficients about CG |
| SM | static margin $\equiv -(\partial C_{m_{\text{CG}}}/\partial\alpha)/(\partial C_L/\partial\alpha)$ |
| $\eta_t = q_{ht}/q_\infty$ | local-to-freestream dynamic-pressure ratio at the htail |
| $c_w$ | wing mean aerodynamic chord, 1.385 m |
| $b$ | wing span, 11.07 m |

## I. Introduction

Distributed-electric propulsion (DEP) over a high-lift wing has opened a new class of *ultra*-STOL (uSTOL) aircraft: very short field performance at modest cruise wing loading, enabled by blowing roughly half the aircraft weight directly across the wing. Electra's EL2 demonstrator and the in-development EL9 hybrid are the most publicly visible examples; both distribute eight to ten propellers along the leading edge, deflecting Fowler-flap augmented section lift

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coefficients well beyond $C_L = 4$. The aerodynamic mechanism by which leading-edge DEP improves take-off and high-lift performance on a commuter-class platform has been quantified in several recent studies [1].

The cost of DEP-augmented blown lift – hidden in a clean configuration drawing – shows up at the empennage. The same propeller wake that creates wing $C_{L,\max}$ pours energetic, deflected flow into the horizontal-tail region. At high thrust setting ($T/L \to 1$) the downwash blanket along the centerline fills the airspace where the stabilizer would operate, and elevator authority collapses precisely when it is needed – at low approach airspeeds, with flaps deployed, and at the highest body angles of attack.

A simple low-mounted horizontal tail behind a continuous-flap wing – the layout adopted by the NASA X-57 "Maxwell" all-electric DEP testbed [2–4] – is known to exhibit stability shortcomings at high thrust setting, and is the natural starting point of any DEP-eSTOL configuration study. The textbook fix is to raise the stabilizer clear of the wake (T-tail, high boom). At the approach speeds and flap settings representative of today's blown-lift uSTOL flight envelopes this fix has proved adequate. This paper, however, targets a more aggressive uSTOL operating frontier – slower approach speeds, stronger blowing, and steeper landing flare attitudes than current designs nominally accommodate – and shows that within that extended envelope *the T-tail fix is necessary but insufficient*: even with a 40%-span, full-chord stabilizer perched at $z = +1.9\,c_w$ and extending from $x = +3.5\,c_w$ to $+4.5\,c_w$ aft of the CG (i.e. $4\,c_w$ aft of the wing leading edge), the configuration remains unstable in landing (and only weakly stable in takeoff). We then walk through a four-step *evolution* that arrives at a much better configuration for this extended envelope: starting from the X-57-style cont-low baseline (Step 0), raise the htail (Step 1, the current wisdom), introduce an inboard flap gap (Step 2), drop the horizontal tail back into the slipstream column to harvest the slipstream dynamic pressure (Step 3, ironically returning to the Step 0 vertical position), and – once the stability headroom is large enough – shorten the tail boom (Step 4).

We perform an unsteady RANS sweep with Flow360 [5–10] on a 10-prop, 165 ft$^2$, 2,600 lb reference uSTOL at ISA 3,658 m. The three phases run at $(V_\infty, \gamma)$ of (45.7 m/s, 0) cruise, (18.0 m/s, +30°) takeoff, and (12.9 m/s, −30°) landing, with sensitivity sweeps centred on baseline $(\alpha, \theta_{ht}, T_{\text{mult}})$ of $(+7, 0, 1)$, $(+8, -5, 16)$, $(+8, -6, 12)$ respectively (BO $T/L$ of 0.05, 0.52, 0.39). Three high-lift section airfoils (Fig. 1) form a Fowler-deployed assembly; the inboard 40% span is optionally a "flap gap." The four configurations form a $2 \times 2$ design matrix in (flap continuity, $z_{ht}$) (Fig. 2). Per combination and phase we run an $\alpha/\theta_{ht}/T/L$ sensitivity sweep, fit local linear sensitivities with stall-point masks, and solve the $3 \times 3$ trim system for the equilibrium flight condition.

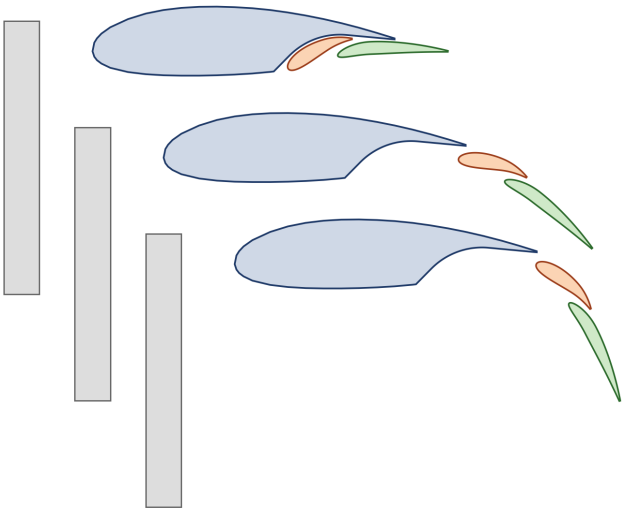


**Fig. 1 Three-element high-lift system in three Fowler kinematic states (top to bottom: stowed, takeoff, landing). Coved LS(1)-0417 main element, NACA 9621 vane, and NACA 6311 aft flap; each successive configuration is shifted downward by $0.3\,c$. The gap-flap variants omit the vane + aft-flap assembly across the central 40% of the wing span.**

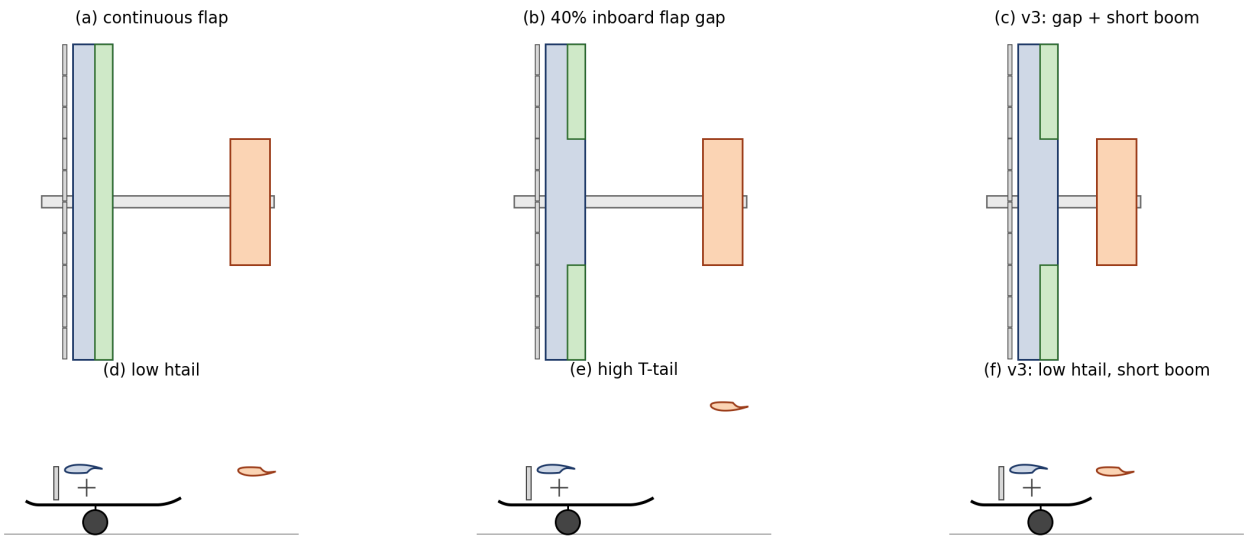



**Fig. 2 Configuration evolution at the cruise (flap-stowed) reference geometry. Top row, top views: (a) continuous flap; (b) 40% inboard flap gap; (c) gap + shortened tail boom (v3, principal proposal). The top views are independent of the horizontal-tail vertical position. Bottom row, side views: (d) low horizontal tail; (e) high T-tail; (f) v3 low htail with the shortened boom. Wing and horizontal-tail sections shown at actual chord scale (clean LS(1)-0417 for the wing; inverted LS(1)-0417 for the htail). Actuator disks are drawn as rectangles (axial-thickness × projected-diameter) so they appear physically correct in both projections. CG is marked "+"; main landing gear shown as a filled circle. The same gear is drawn in all three side views to highlight the rotational advantage of the short-boom v3: at high landing-flare attitudes the aircraft pitches up around the main-gear contact point, and the shorter-arm htail in (f) clears the ground at flare attitudes where the long-boom configurations (d, e) would strike.**

## II. Configurational Evolution

**Note on grid uniformity.** The CFD sweeps that feed Table 1 and the downwash slices (Figs. 4–12) were carried out at each configuration's own equilibrium trim point rather than on a single shared $(T/L, \theta_{ht})$ grid: the landing slices sit at $T/L \approx 0.60$ for cont-low, 0.39 for cont-high and gap-high, and at the respective BO values for gap-low and v3. Because slipstream downwash scales with $T/L$, cross-configuration magnitudes and SM deltas inherit a thrust-dependent bias of unknown sign; the *directions* of the effects we report (gap-induced upwash, slipstream $q$-boost, short-boom moment-arm penalty) are robust to this bias, but quantitative cross-configuration deltas should be read as indicative. The full paper will rerun every configuration on a common $(T/L, \theta_{ht})$ grid and report calibrated deltas. The downwash plots also use a body-fixed slice (perpendicular to body $x$ at $x_b = +4.4$ m) with Flow360's freestream pinned along world $+x$ and the airframe rotated by $\theta$ via the `ac_pitch` rotation cylinder, so the plotted $\epsilon = \arctan(-w_{\text{world}}/u_{\text{world}})$ is measured against the freestream direction and the htail dotted line sits at constant $z_{\text{body}}$ across all $\alpha$ panels.

### A. Step 0 – The Naive Low-Tail Baseline (cont-low)

The simplest DEP-eSTOL configuration places a conventional low-mounted horizontal tail behind a continuous-flap wing. This is the layout of the NASA X-57 "Maxwell" all-electric DEP testbed [2–4], the most publicly visible example of which is known to have experienced flight-readiness challenges that contributed to the program's 2023 closure. For our 10-prop, full-span distributed array this layout is fine in unflapped cruise – a clean, lightly-loaded wing and a tail outside the (small) prop wake gives a textbook SM = +46.7% MAC. The failure modes the X-57 community anticipated emerge in the flapped phases: Figure 3 shows the sensitivity sweeps across the three phases. In takeoff and landing the horizontal tail sits inside the wing+flap downwash and is hit by the wakes of the inboard propellers; both contribute to a destabilising downwash gradient that erodes the tail's restoring moment. The static-margin column of Table 1 reads SM = +2.9% in takeoff (barely stable) and SM = −19.5% in landing (strongly unstable); cont-low sets the benchmark against which the subsequent steps are measured.

The downwash field at the htail station makes the static-margin numbers physical. Figure 4 shows the wind-frame

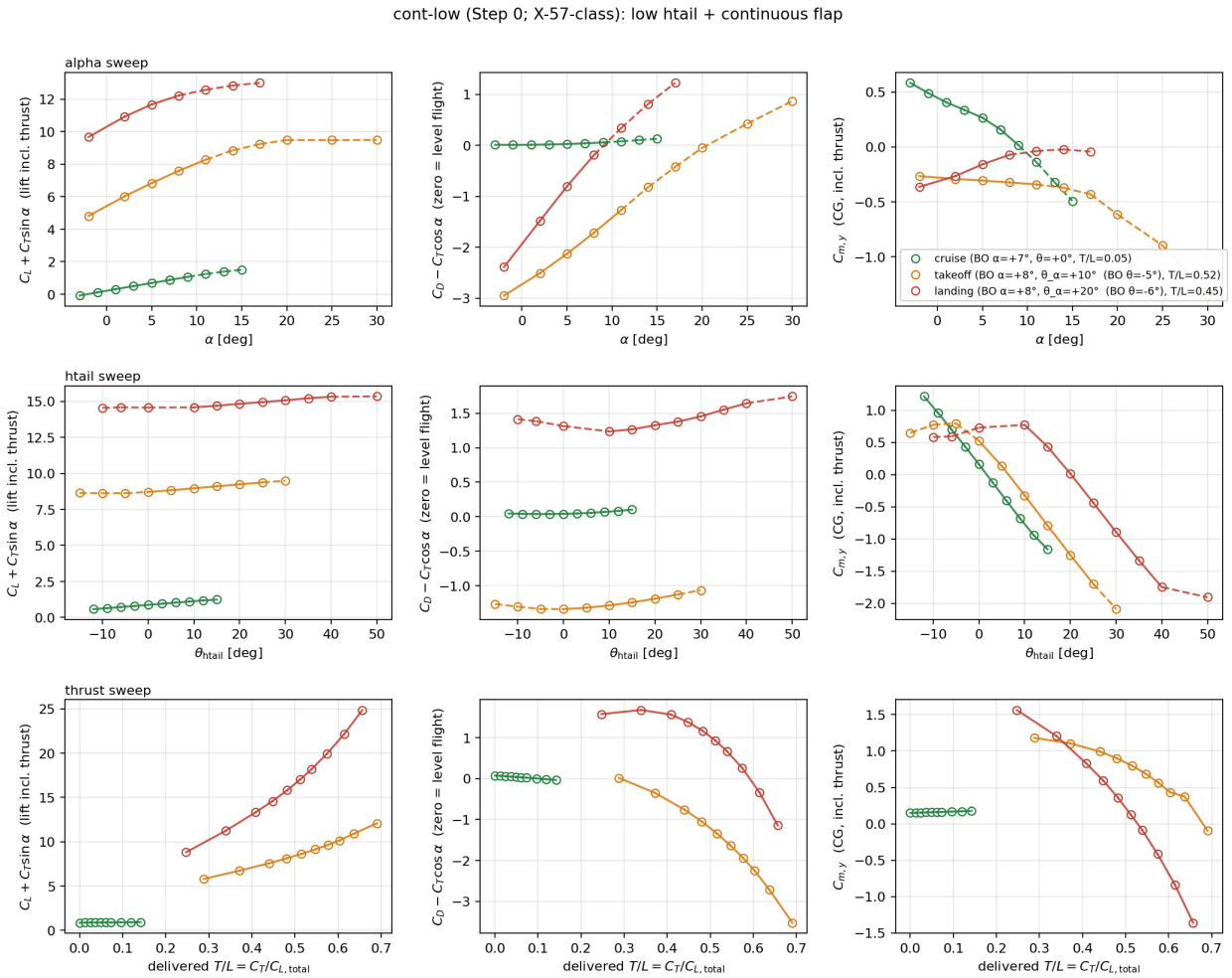



**Fig. 3 Sensitivity sweeps for the cont-low Step 0 baseline (the X-57-class layout) with all three flight phases overlaid on a common set of axes: cruise (green), takeoff (orange), landing (red). Lift and drag columns include the streamwise thrust contribution: $C_{L,\text{total}} = C_L + C_T \sin\alpha$ and $C_{D,\text{total}} = C_D - C_T \cos\alpha$, so that $C_{D,\text{total}} = 0$ marks level flight (climb when negative, descent when positive). The $C_{m_{\text{CG}}}$-vs-$\alpha$ slope in the third column is positive in both takeoff and landing, indicating instability; cruise on the clean wing is comfortably stable. Cont-low sets the benchmark against which the subsequent steps are measured.**

downwash angle $\epsilon = \tan^{-1}(-w_{\text{world}}/u_{\text{world}})$ on a body-fixed yz-slice 0.45 m upstream of the htail leading edge at landing trim, for two angles of attack ($\alpha = +5°$ and $+17°$). Reading $\epsilon$ along the htail dotted line ($z_{\text{body}} = +0.554$ m): at $\alpha = +5°$ the htail sits in $\epsilon \approx +45°$ downwash, and at $\alpha = +17°$ it sits in $\epsilon \approx +55°$. A 12° increase in body $\alpha$ thus produces a $\sim 10°$ increase in downwash at the htail, so the effective $\alpha$ seen by the htail rises by only $\sim 2°$ – the htail is almost completely shielded from the body's pitch motion, and the SM = −19.5% landing entry of Table 1 follows directly. An upward gust on the airframe is intercepted by the downwash blanket before it can produce the htail restoring moment a low-DEP-loading aircraft would rely on.

### B. Step 1 – The Current Wisdom: T-Tail with Continuous Flap (cont-high)

The traditional remedy is to lift the stabiliser clear of the wake: cont-high replaces the low htail with a T-tail at $z = +1.9\,c_w$, recovering elevator effectiveness through a reduced downwash gradient. Figure 5 shows the sensitivity sweeps across the three phases.

Three uSTOL-relevant observations follow:

1) The T-tail buys back stability in cruise but *not* in landing – exactly the phase that stresses empennage authority

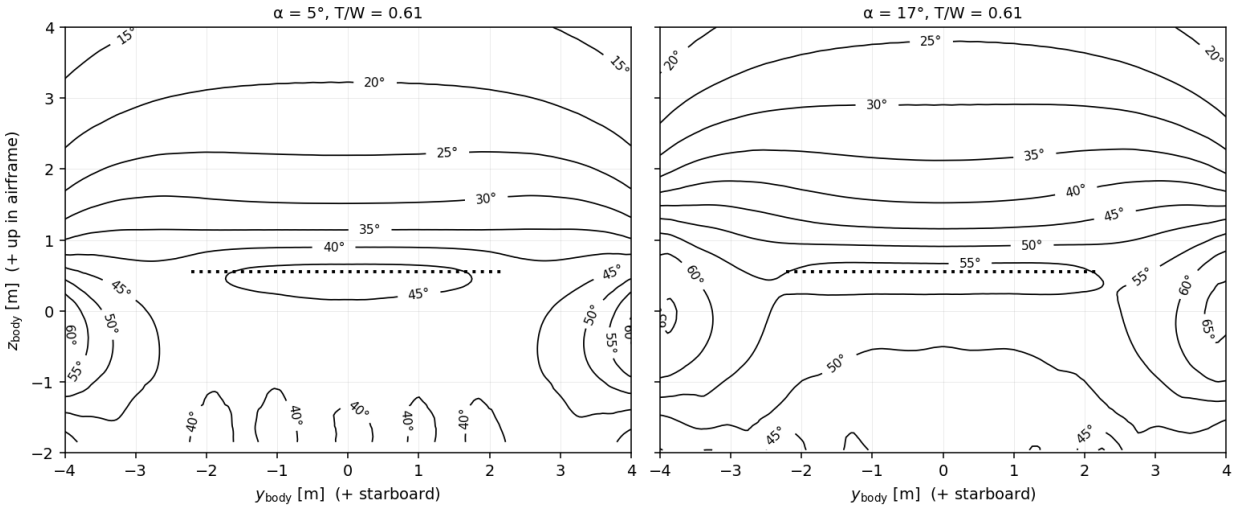



**Fig. 4 Wind-frame downwash $\epsilon$ at the cont-low htail station ($x_b$ = htail LE $-0.45$ m, landing trim, $T/L \approx 0.60$), at $\alpha = +5°$ (left) and $\alpha = +17°$ (right). Labeled contour lines are degrees of $\epsilon$; the dotted black line marks the htail span at its body-z station. The htail sits inside a deep downwash column ($\epsilon \gtrsim +20°$) – the geometric mechanism behind the cont-low landing instability.**

most.

2) Elevator authority (coefficient slope) is similar in cruise and landing for cont-high ($|\partial C_{m_{\mathrm{CG}}}/\partial\theta_{ht}| \approx 0.087$/deg cruise, 0.078/deg landing – only $\sim 10\%$ lower at landing). In *dimensional* terms, however, the moment authority $q \cdot S \cdot c \cdot |dC_m/d\theta_{ht}|$ scales with $q$, and the landing $q$ is $(V_{\mathrm{land}}/V_{\mathrm{cruise}})^2 = (12.86/45.72)^2 \approx 8\%$ of cruise. Despite the stabiliser being 40% of the wingspan and full main-wing chord (large by any conventional metric), the marginal trim deflections sit close to the empirical htail stall boundary (+12° takeoff, +20° landing).
3) In the negative-SM landing regime, the htail operates at high lift to trim the wing+flap moment. If the htail stalls in this state – e.g. during a flare gust upset – $|\partial C_{m_{\mathrm{CG}}}/\partial\theta_{ht}|$ collapses to zero, the wing+flap nose-down moment is unopposed, and the aircraft enters a nose-diver attitude with no static-recovery path. This is a catastrophic failure mode at landing altitude.

The textbook fix is necessary, but at the extended-envelope landing condition this paper targets it does not deliver a credible safety margin. At today's published Electra-class approach speeds the same configuration would be acceptable; the analysis above is best understood as describing what happens when one tries to push a conventional empennage beyond its current operating frontier.

The downwash field shows the partial nature of the fix. Figure 6 shows wind-frame $\epsilon$ on the same upstream-of-htail yz-slice, with the htail now at $z_{\mathrm{body}} = +2.63$ m. Reading the dotted line, the htail sits in $\epsilon \approx +30°$ at $\alpha = +5°$ and $\epsilon \approx +37°$ at $\alpha = +17°$. A 12° body-$\alpha$ increase now produces only a $\sim 7°$ downwash increase at the htail, so the htail's effective $\alpha$ rises by $\sim$ 4–5° – a meaningful (but mild) improvement over cont-low's $\sim 2°$. The T-tail clears the densest core of the wing-flap downwash but only blunts, rather than reverses, the landing-destabilising gradient: the residual $d\epsilon/d\alpha > 0$ at the htail is the geometric explanation for the still-unstable landing entry in Table 1.

### C. Step 2 – Introducing the Flap Gap (gap-high)

The first step of the evolution is to remove the Fowler vane and aft-flap assembly from the central 40% of the wing span; the main wing element remains continuous (this is a *flap* gap, not a wing gap), and the inboard and outboard flap segments are unchanged.

The gap costs lie in four predictable channels (reported qualitatively here): (i) the missing inboard flap drops landing $C_{L,\max}$ relative to the continuous-flap configurations, smaller in takeoff where the flap is less deflected; (ii) the unblown inboard wing loses slipstream lift augmentation, raising the trimmed approach speed at fixed weight and thrust – a real cost for a slow-approach uSTOL; (iii) the de-blown inboard span sharpens the lift drop-off at the gap edges and adds two streamwise vortices' worth of induced drag; (iv) the lower local $C_{L,\max}$ of the unblown inboard wing biases incipient stall inboard – the opposite of a benign washout-twisted GA pattern – which needs explicit verification in the full paper.

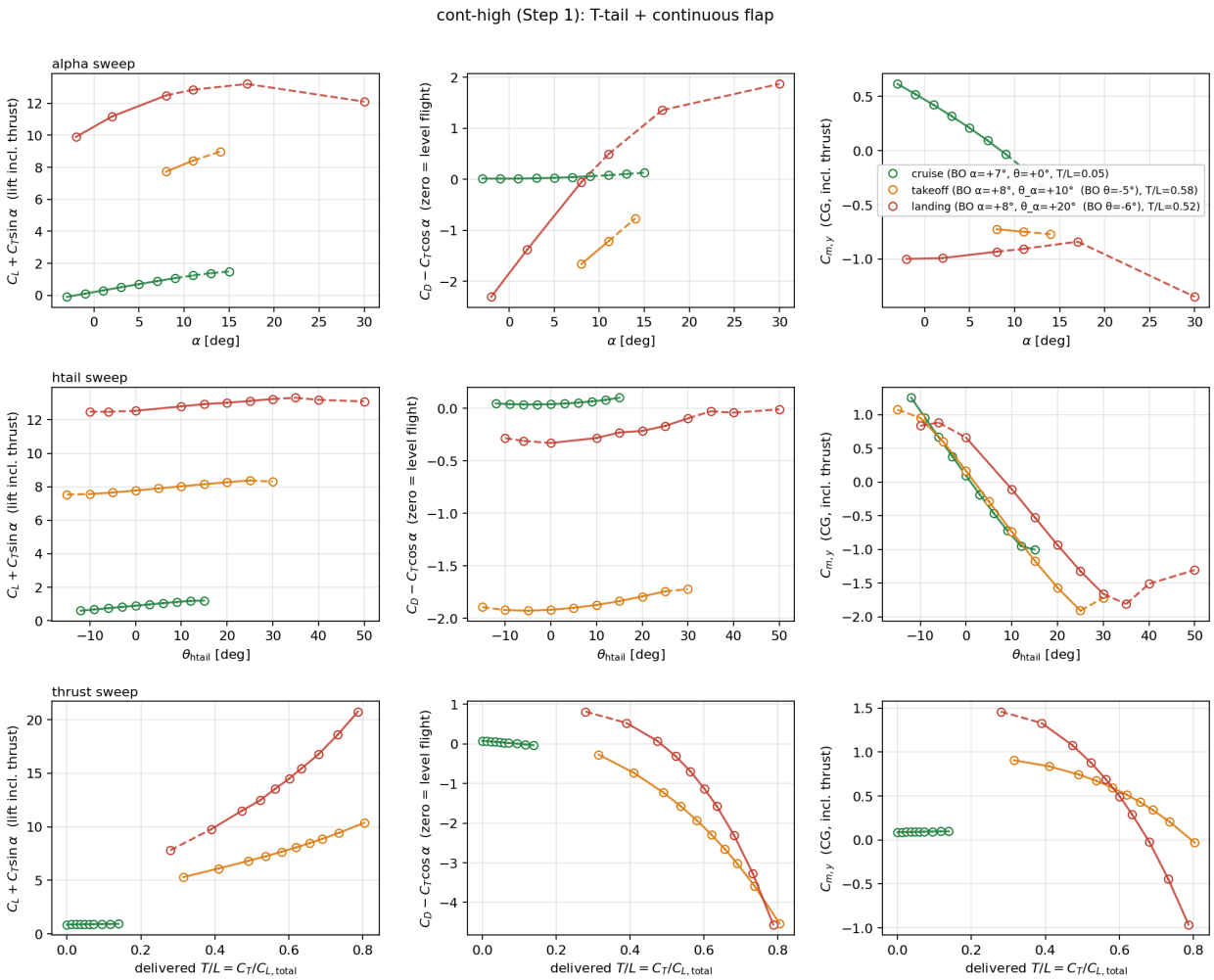



**Fig. 5 Sensitivity sweeps for the cont-high T-tail baseline (Step 1), three flight phases on common axes: cruise (green), takeoff (orange), landing (red). Rows are the three sensitivity sweeps ($\alpha$, $\theta_{ht}$, $T/L$); the first two columns are $C_{L,\text{total}} = C_L + C_T \sin\alpha$ and $C_{D,\text{total}} = C_D - C_T \cos\alpha$, so $C_{D,\text{total}} = 0$ corresponds to level flight. Solid line segments lie inside the linear-fit range; dashed segments lie outside. See Table 1 for the static-margin entries.**

The small cruise drag improvement (cleaner unflapped central section vs. slightly draggy stowed flap) is real but tiny.

The benefit is the htail-side mechanism evident in Fig. 8. Two counter-rotating streamwise vortices, shed at the gap edges and clearly visible as $\epsilon$-extrema centred near $y = \pm 2.2$ m, $z_{\text{body}} \approx -1$ m, roll the inboard prop wake inward and replace the continuous-flap downwash column with a centerline *upwash* core. Reading along the htail dotted line at $z_{\text{body}} = +2.63$ m: $\epsilon$ rises from $\sim 0°$ at $\alpha = +5°$ to only $\sim +5°$ at $\alpha = +17°$. A 12° body-$\alpha$ increase produces just a $\sim 5°$ downwash increase at the htail, so the htail's effective $\alpha$ rises by $\sim 7°$ – distinctly more than cont-high's 4–5° and far above cont-low's $\sim 2°$. By the htail-contribution sign rule this further *stabilises* $\partial C_{m_{\text{CG}}}/\partial\alpha$, taking landing SM from negative to positive and adding margin in takeoff; the gap is mildly destabilising in cruise where the unflapped wing sheds no central downwash to invert (Table 1).

### D. Step 3 – Ironically, Returning the htail to the Low Position (gap-low)

With the gap providing the stability fix that the high tail alone could not, the design optimisation flips. We are now free to bring the horizontal tail back *down*. Ironically this places the stabiliser at the same vertical position as in the Step 0 naive baseline – but now, with the upwash column from the flap gap providing the stability backbone, the low position is no longer a liability. Instead it is a strong positive: at $z = +0.4\,c_w$ the htail sits inside the merged 10-prop slipstream

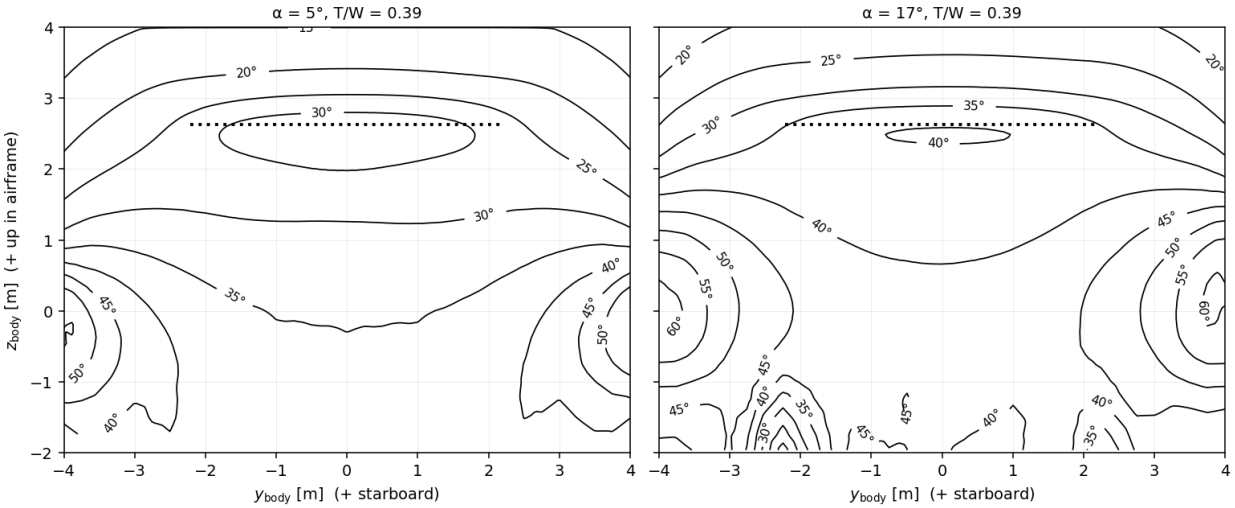



**Fig. 6 Wind-frame downwash $\epsilon$ at the cont-high htail station ($T/L \approx 0.39$), $\alpha = +5°$ and $+17°$; same slice and conventions as Fig. 4. Lifting the stabiliser to the T position clears the densest core of the downwash column, but a wide $\epsilon > +15°$ band still envelops the htail span at high $\alpha$. The residual $d\epsilon/d\alpha > 0$ at the htail station is the geometric explanation for the still-unstable landing entry in Table 1.**

column. Locally the htail sees the slipstream velocity rather than the freestream; at landing thrust $V_{\text{slip}}/V_\infty \approx 2$–$3$, giving a 4–9× dynamic-pressure boost. Both $\partial C_{m_{\text{CG}}}/\partial\alpha$ (stability) and $\partial C_{m_{\text{CG}}}/\partial\theta_{ht}$ (elevator authority) scale with local $q$.

**Table 1 Static margin by configuration and phase, from refreshed CFD sensitivity fits with stalled-point masks applied. cont-low and cont-high takeoff and landing fits use the $\alpha$-sweep at $\theta_{ht} = +10°$ (takeoff) / $+20°$ (landing) to keep the htail in its linear regime; the BO $\alpha$-sweep at the much lower $\theta_{ht} = -5°/-6°$ was contaminated by htail stall at high $\alpha$ and produced significantly more negative apparent static margins (e.g. cont-low landing $-29.7\%$ vs the unstalled value $-19.5\%$). Negative SM indicates instability.**

| Configuration | Cruise SM | Takeoff SM | Landing SM |
|---|---|---|---|
| **cont-low (Step 0; X-57-class)** | +46.7% | +2.9% | −19.5% |
| cont-high (Step 1; T-tail, continuous flap) | +55.4% | +4.8% | −4.6% |
| gap-high (Step 2; T-tail, 40% flap gap) | +51.5% | +12.1% | +6.4% |
| **gap-low (Step 3; gap + slipstream)** | **+48.6%** | **+73.1%** | **+91.6%** |
| **v3 (Step 4; gap-low + short boom)** | **+14.9%** | **+19.0%** | **+21.4%** |

The gap-low headline numbers (Table 1) are strongly stable in every phase. Elevator authority roughly doubles relative to cont-high in takeoff and landing, exactly as expected for a $\sim 2\times$ velocity boost on the htail.

The downwash field at the gap-low htail station, Figure 10, combines the two mechanisms. The htail is back at low $z_{\text{body}}$ – the same geometric station that buried it in cont-low – but the local flow is qualitatively inverted: the centerline gap-upwash column still reaches the lower htail position, while the slipstream column accelerates local $u_{\text{world}}$ to roughly twice freestream (visible as compressed iso-$\epsilon$ lines relative to cont-low at matched $\alpha$). Reading along the htail dotted line, the $d\epsilon/d\alpha$ benefit at the low station is somewhat weaker than at the gap-high T-tail position (the inboard gap-edge vortices are farther from the low htail vertically), but the slipstream-doubled local dynamic pressure dominates the htail-contribution arithmetic: the htail's restoring moment per degree of effective $\alpha$ scales with $\eta_t = q_{ht}/q_\infty \approx 4$, and that is the mechanism behind the Table 1 jump to strongly stable in every flapped phase. Both pieces – the gap-induced $d\epsilon/d\alpha$ flip and the slipstream $q$-boost – carry over to v3 (Step 4).

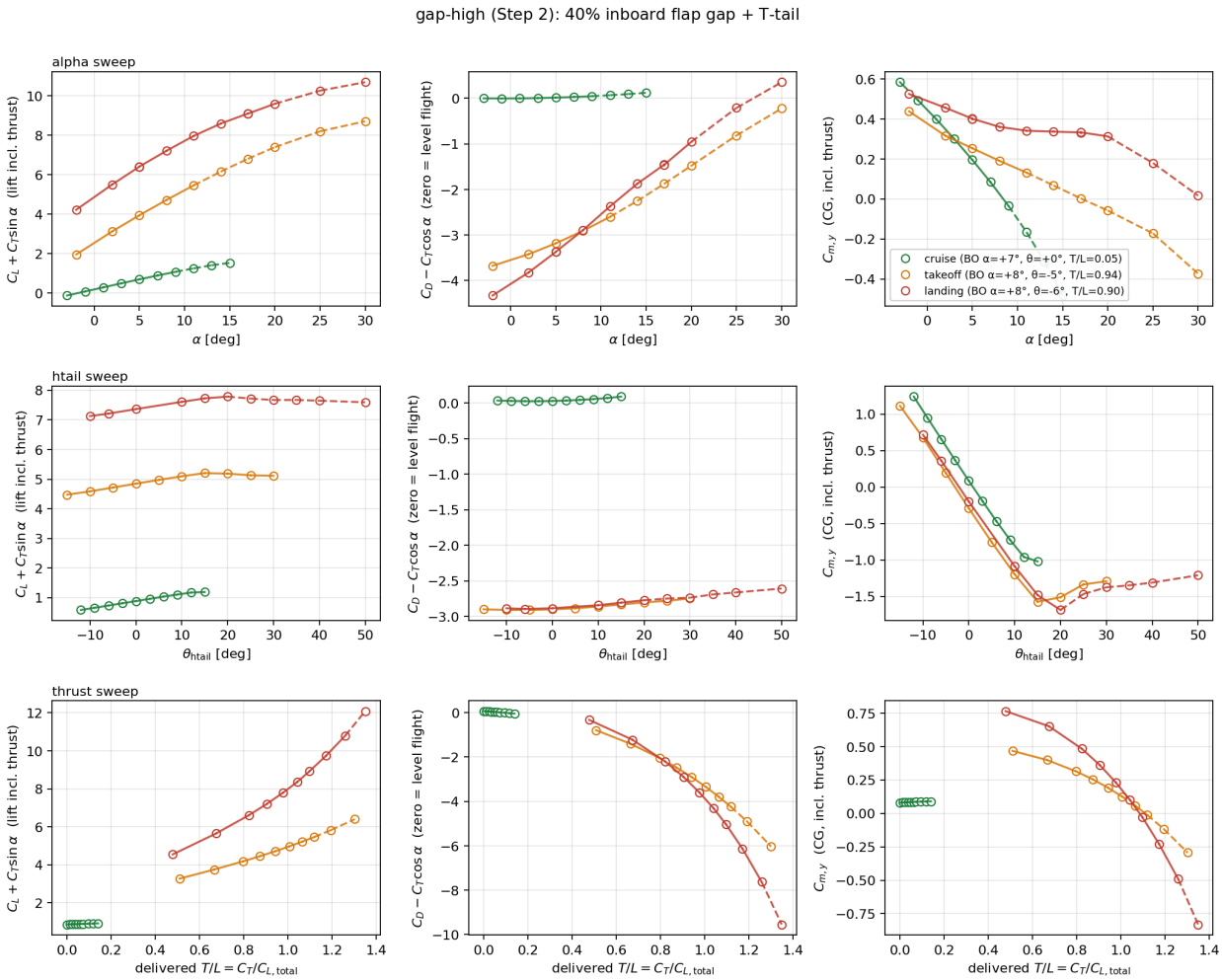



**Fig. 7 Sensitivity sweeps for the gap-high configuration (Step 2). Same axes, color encoding, and $C_{L,\text{total}}/C_{D,\text{total}}$ convention as Fig. 5. Compared with the cont-high baseline, the flap gap stabilises takeoff and landing: takeoff goes from weakly stable to moderately stable, and landing flips from mildly unstable to weakly stable – numbers in Table 1. Mechanism: streamwise vortices at the flap-segment edges roll the inboard prop wake *inboard* along the wing-trailing-edge gap, producing a coherent upwash column that impinges on the lower surface of the horizontal tail; local $d\epsilon/d\alpha$ at the htail station effectively flips sign relative to the continuous-flap configuration, steepening the htail's restoring moment.**

### E. Step 4 – Shortening the Tail Boom (v3)

The gap-low landing SM = +91.6% MAC is unusually stiff for a piloted aircraft – roughly an order of magnitude above a typical GA tail-volume margin. This headroom is real, and inviting to spend. Three structural levers reduce it back toward a more conventional range while preserving the gap-induced stability mechanism: (i) reduce the htail span, (ii) reduce the htail chord, or (iii) shorten the tail boom. Each cuts the htail's contribution to $\partial C_{m_{CG}}/\partial\alpha$, and each carries a different secondary cost. Span reduction is the most risky: it shrinks the htail's exposure to the slipstream-doubled local dynamic pressure, and – more importantly – it concentrates the remaining htail span over fewer of the four inboard centerline blowers, so loss of one inboard propeller punches a larger fractional dead-air hole in the htail's effective $q$. Chord reduction trades nominal-flight stability cleanly against weight and structural depth and is the most benign on prop-out scenarios. Boom shortening pays in pitch-stability arm-length but *gains* in landing-flare clearance and pilot-input sensitivity (see Sec. III.A and the flare-geometry argument in Sec. 3). All three are worth investigating; this abstract restricts attention to boom shortening (v3 below) because it is the most architecturally distinctive and because the flare-geometry consequences are unique to that lever. A practical preliminary design would likely combine boom

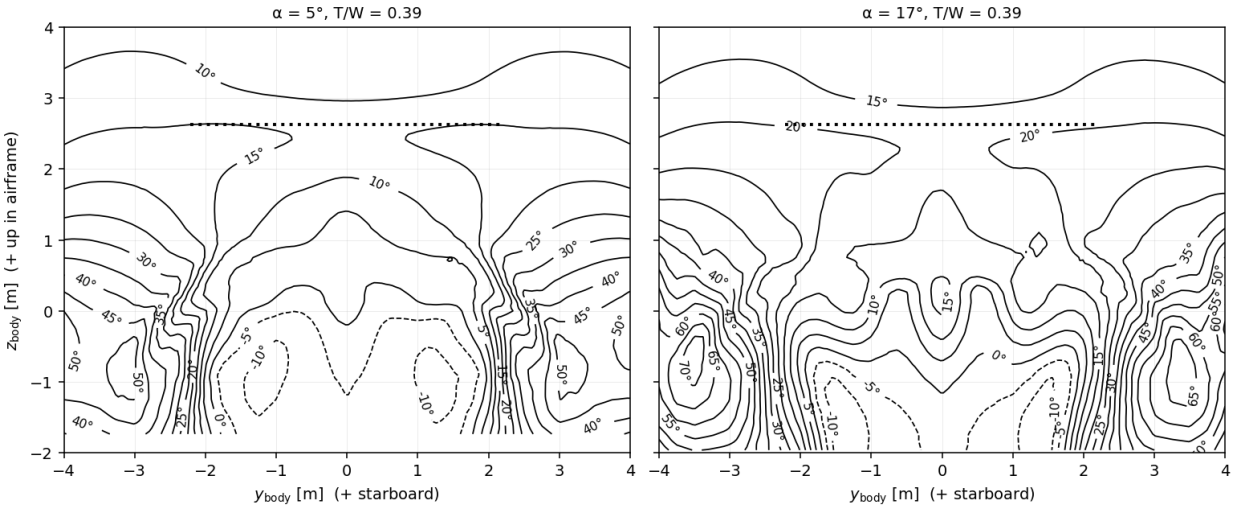



**Fig. 8 Wind-frame downwash $\epsilon$ at the gap-high htail station ($T/L \approx 0.39$), $\alpha = +5°$ and $+17°$; same slice and conventions as Fig. 4. The flap gap replaces the continuous-flap downwash column with a centerline *upwash* column ($\epsilon < 0$) flanked by two flap-edge vortices. The upwash strengthens with $\alpha$ – the $d\epsilon/d\alpha < 0$ that flips the htail contribution to $\partial C_{m_{\mathrm{CG}}}/\partial\alpha$ from destabilising to stabilising, consistent with the landing** SM **flip from negative to positive in Table 1.**

shortening with a modest chord reduction; the full paper will revisit the design matrix on that basis.

v3 retains the gap-low planform but moves the htail aerodynamic centre from $+3.75\,c_w$ to $+2.0\,c_w$ aft of the CG (from $\sim 4\,c$ aft of the wing LE to $\sim 2\,c$ aft of the wing TE), holding $z = +0.4\,c_w$ to stay inside the slipstream column. This is the principal configurational contribution of this paper and is analysed in the next section.

## III. Analysis of the v3 Configuration

v3 halves the htail moment arm ($k = L_{ht}^{(v3)}/L_{ht}^{(v2)} = 2.0/3.75 \approx 0.53$) while retaining the gap-low planform and high-lift system. The htail's $\partial C_{m_{\mathrm{CG}}}/\partial\alpha$ and $\partial C_{m_{\mathrm{CG}}}/\partial\theta_{ht}$ scale linearly with the moment arm; wing and flap contributions do not. The v3 CFD campaign ($3 \times (1 + 19)$ cases) is in flight with partial coverage (landing $7/20$, takeoff $5/20$, cruise queueing). BO trims sit at the low-slow-pass frontier ($\alpha = +15°, \theta_{ht} = -5°$ takeoff; $\alpha = +25°, \theta_{ht} = +5°$ landing) to probe the short boom under the most demanding tail-authority condition (Fig. 11).

Figure 12 shows $\epsilon$ on the v3-specific yz-slice ($x_b$ = htail LE $-0.23$ m). The gap-upwash and slipstream-acceleration features of gap-low are preserved at the v3 station; the htail span sits inside the upwashed, accelerated column. This is the geometric basis for v3 retaining usable stability and elevator authority despite the $\sim 47\%$ moment-arm reduction.

### A. Trimmable flight envelope

To read off what the configuration can *do*, we march through an $(\alpha, T/L)$ grid: at each point we pick $\theta_{ht}$ to trim the moment, evaluate $C_L$, $C_D$, $C_{T,\mathrm{del}}$ from local linear sensitivities, and solve the Newtonian force-balance for equilibrium $V$ and $\gamma$. The table is generated from the *partial v3 short-boom* CFD sensitivities currently on hand (`post/v3/plot_*_sensitivities.py`), centred on the v3 BO trim points ($\alpha = +7/+15/+25°$ in cruise/takeoff/landing) and spanning the low-slow-pass envelope. Final numbers refresh when the remaining v3 sweep points land.

In cruise ($\alpha \in [+5, +9]°$) the aircraft trims with $\theta_{ht} \in [+3.6, +5.3]°$, modest deflections inside the htail's unstalled range, with $|\gamma| \lesssim 3°$ and $V \in [75, 92]$ kt across the cruise-thrust grid. In takeoff ($\alpha \in [+9, +15]°$, $T/L \in [0.37, 0.78]$) trim $\theta_{ht}$ is mildly negative everywhere ($\leq 0°$, inside the $+12°$ stall margin); $\gamma$ runs from a mild $+3$–$+6°$ climb at $T/L \approx 0.37$ to a steep $+22°$ blown-lift climb at $T/L \approx 0.77$, with $V \in [23, 43]$ kt throughout. In landing ($\alpha \in [+15, +25]°$) approach is steep at low thrust ($\gamma \in [-11, -16]°$ at $T/L \approx 0.22$), shallower at mid-thrust ($-7$ to $-10°$), and near-level at high-thrust ($\gamma \approx -1°$, a power-on go-around margin); $V \in [26, 35]$ kt and the maximum

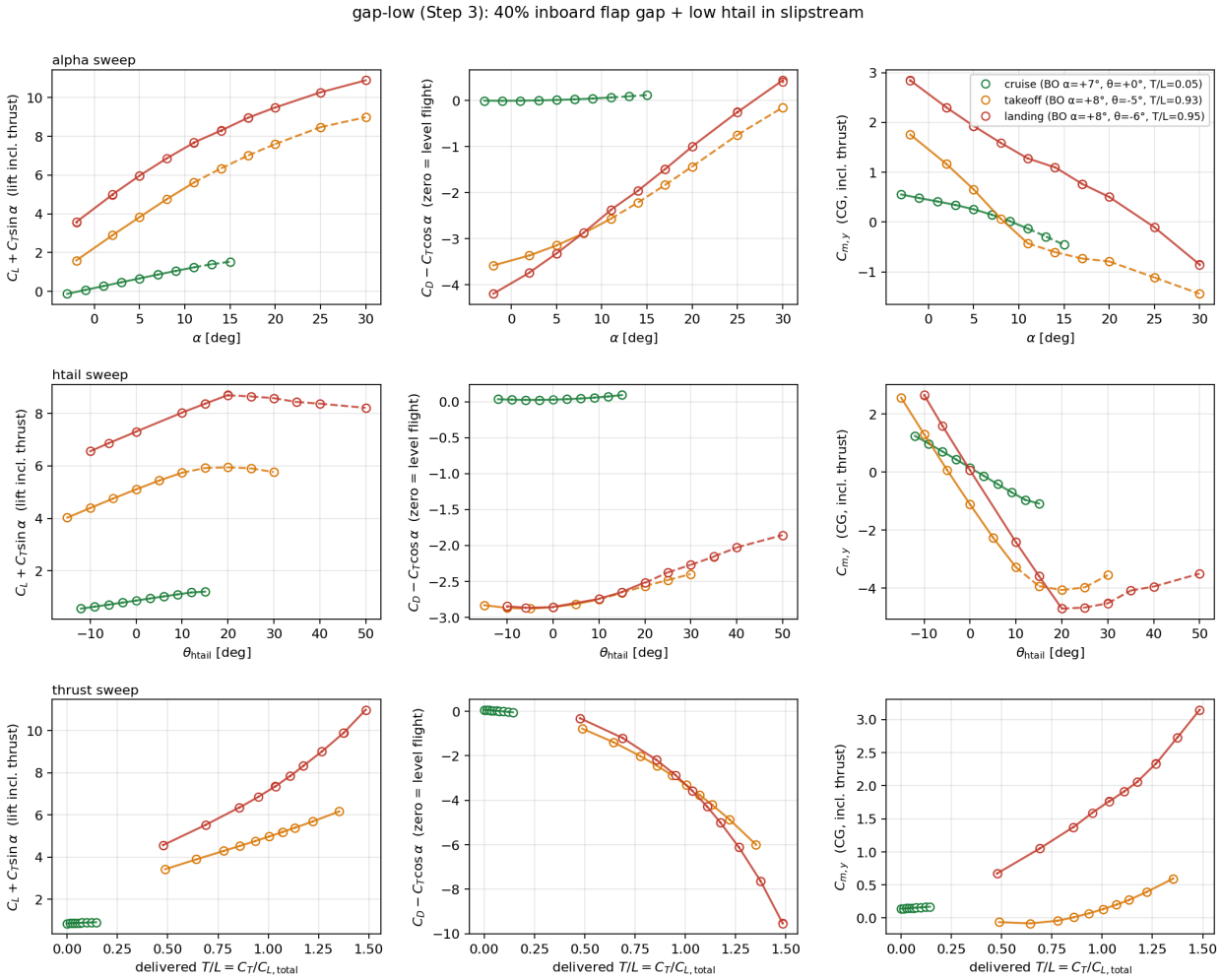



**Fig. 9 Sensitivity sweeps for the gap-low configuration (Step 3). Same axes, color encoding, and $C_{L,\text{total}}$/$C_{D,\text{total}}$ convention as Fig. 5. All three phases of gap-low are strongly stable: the $C_{m_{\text{CG}}}$-vs-$\alpha$ slopes (third column) are sharply negative and the elevator- authority slopes ($\partial C_{m_{\text{CG}}}/\partial\theta_{ht}$, middle row, third column) are roughly doubled relative to cont-high. The htail is in the prop slipstream and benefits from a 4–9× dynamic-pressure boost. Reading $C_{D,\text{total}}$ in the thrust-sweep row (bottom-middle) gives a direct visualisation of the level-flight thrust setting at each phase: the $C_{D,\text{total}} = 0$ crossing identifies the trimmed level-flight thrust setting at that phase's airspeed.**

trim magnitude $|\theta_{ht}| = 6.1°$ sits comfortably inside the +20° landing stall margin. The v3 design target – $\alpha \approx +25°$, $T/L \approx 0.4$, $\gamma \approx -10°$, $V \approx 29$ kt – sits squarely inside the grid.

### B. Slipstream coverage at the shorter arm

The merged 10-prop slipstream column spreads from $z \in [-0.3, +0.6]\, c_w$ at the v2 htail station ($4.4\, c_w$ downstream of the disks) to a slightly tighter $z \in [-0.2, +0.55]\, c_w$ at the v3 station ($2.7\, c_w$ downstream). The v3 htail at $z = +0.4\, c_w$ sits comfortably inside this band; moving it up by more than $\sim 0.1\, c_w$ would risk exiting the upper edge, which is why v3 retains the v2 low-tail vertical position.

### C. Geometric benefits at landing flare

Rotating about the main-gear contact at $(x_g, z_g) = (+0.50, -1.00)\, c_w$ maps a body-frame point $(x, z)$ to $z' = z_g - (x - x_g)\sin\theta + (z - z_g)\cos\theta$. For the htail TE at $z = +0.4\, c_w$ and a 30° flare attitude this gives $z'_{TE} = -1.79\, c_w$

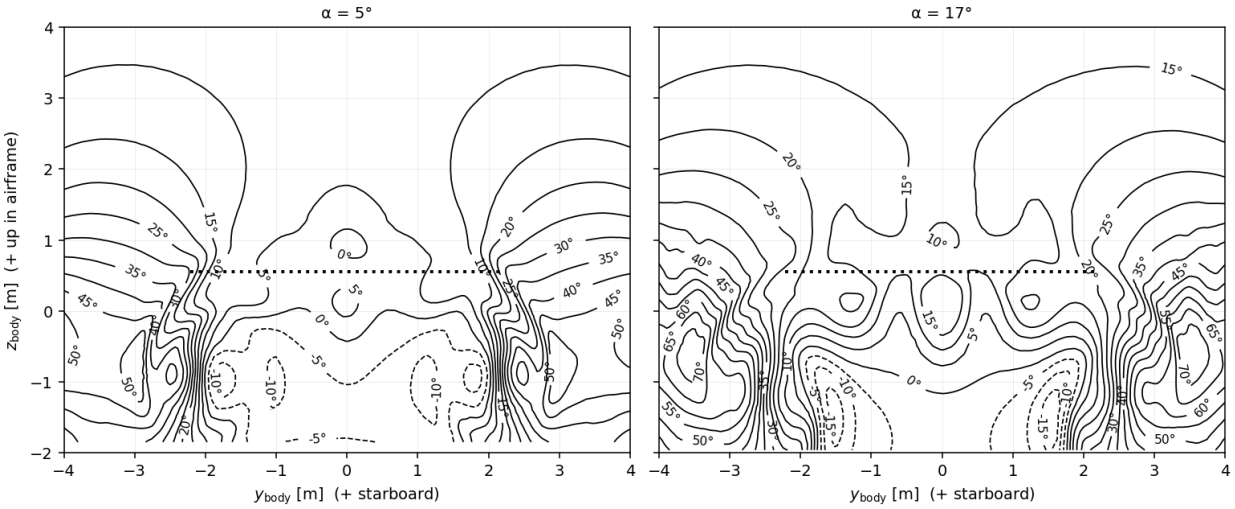



**Fig. 10 Wind-frame downwash $\epsilon$ at the gap-low htail station, $\alpha = +5°$ and $+17°$; same slice and conventions as Fig. 4. The low htail position is recovered ironically: the gap-upwash column reaches the htail span, and the slipstream column lifts local $u_{\text{world}}$ to $\approx 2\,V_\infty$. Combined, these give the large-margin landing** SM **and the $\approx 2\times$ elevator-authority boost in Table 1.**

for v2 ($x_{TE} = +4.5\,c_w$, $0.79\,c_w$ below ground – tail strike) versus $z'_{TE} = -0.79\,c_w$ for v3 ($x_{TE} = +2.5\,c_w$, $0.21\,c_w$ of clearance). At a conventional 17° flare v2 is still marginal ($0.17\,c_w$ below ground) while v3 is easy ($0.75\,c_w$ above). The v3 short boom therefore supports the uSTOL low-slow-pass landing ($\alpha \approx +30°$ at $\gamma = 0$ in ground effect, $V \lesssim 20$ kt) – one row beyond the bottom of Table 2, a target the in-progress v3 sweep extension is configured to probe directly.

### D. Cautions and CFD validation plan

The partial v3 CFD sweep reported here (Table 1) gives measured static margins of +15/+19/+21% MAC across cruise/takeoff/landing – significantly smaller than the simple-arm-scaling linear projection (~ 0.53× of the v2 gap-low slopes would have predicted +26/+39/+49% MAC). The shortfall is consistent with the known steepening of the wing-wake downwash gradient at the htail station as the htail moves closer to the wing: for flapped-wing configurations this effect typically reduces the effective $\partial C_{m_{\text{CG}}}/\partial\alpha$ contribution of the htail by 15–25% beyond simple arm-scaling, and an additional reduction of similar magnitude is plausible from the short boom changing the local downwash field structure. All three v3 phases remain strongly stable at these reduced margins. Completion of the v3 CFD campaign (full 1 + 19 sweep per phase) is in flight and will sharpen these values for the full paper.

**Asymmetric prop-out under the short boom.** The 47% shorter boom also shortens the vertical-tail yaw arm, weakening rudder recovery from asymmetric thrust loss. Four of the ten disks sit on the inboard gap region and blow directly toward the empennage, so the rudder stays inside an energised slipstream after one or two *outboard* disks fail. What the geometry does *not* protect against is a simultaneous loss of one of these four centerline blowers at low airspeed: the gap-low/v3 upwash column that anchors pitch stability is itself partially de-energised, and the short boom no longer has the lever arm to recover. We anticipate v3 will require an eSTOL height-velocity ("dead man's curve") exclusion region in $(V, h_{\text{AGL}})$ – not a defect unique to v3, but amplified enough by the short boom that it must be characterised rather than assumed small.

**Actuator-disk methodology limitations.** The uniform-pressure-jump actuator disk captures the time-averaged axial thrust and slipstream tube but omits prop swirl and discrete helical tip vortices. Swirl is the concern here: the gap-low/v3 story relies on a *symmetric* centerline upwash column from the flap-gap edge vortices, and an inboard-prop swirl component of the same handedness as one edge vortex (and opposite the other) could deflect the column off-centre. A swirl-augmented disk or a rotor-resolving sliding-mesh spot-check at the v3 landing trim point (Sec. V) is needed to bound the error; the asymmetric prop-out cases above are the most likely regime for actuator-disk and rotor-resolving

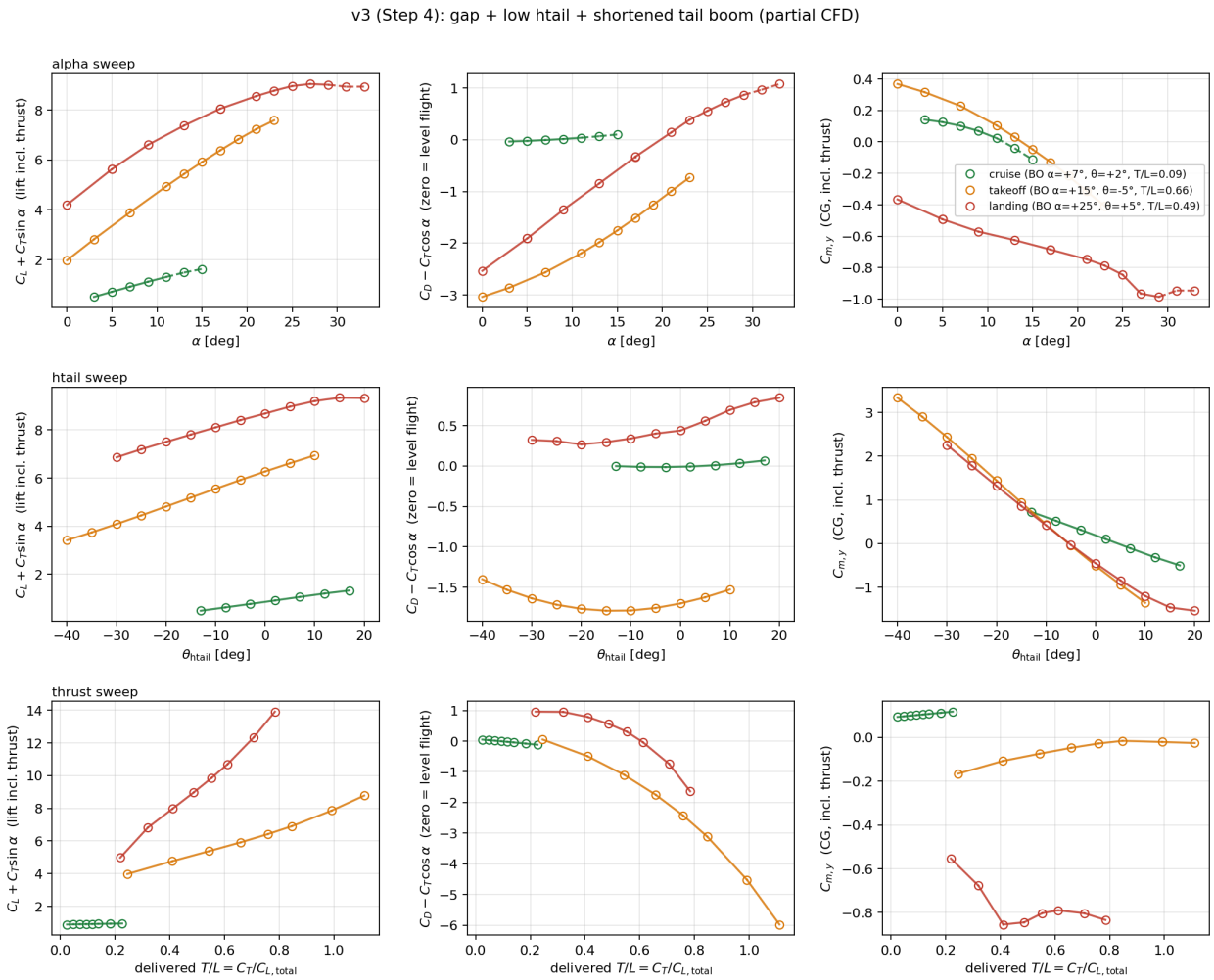



**Fig. 11 v3 (Step 4) sensitivity sweeps; cruise/takeoff/landing rows show the points completed at submission.**

predictions to diverge.

## IV. CFD-Based Dynamic Maneuver Simulation

Steady CFD with the rigid body fixed underpins Secs. II–III.A, but the eventual target is a *dynamic* landing flare with the ground plane. We have built and exercised a CFD-coupled 3-DOF longitudinal framework on a v3 cruise phugoid as a stepping stone so the flare simulation can be set up with confidence.

### A. Method: nested rotation zones + UDD 3-DOF

Flow360 couples CFD to user dynamics through (i) rotation volume zones (sliding-mesh interfaces about an axis) and (ii) the User-Defined-Dynamics (UDD) block, which computes a state from integrated forces and moments and feeds the rotation-zone angles via `FromUserDefinedDynamics`. We chain four nested rotation cylinders in a robot-arm topology (Fig. 13): an outer 2-link arm (*shoulder* + *elbow*, $L_1 = L_2 = 750\,c_w$) translates the airframe through the workspace; inner cylinders rotate the airframe about its CG and the htail about its pivot. Each physical step the UDD reads the airframe wall force $(F_x, F_z, M_y)$, integrates $\dot{V}_x = F_x/m + T_{\text{disk}}/m$, $\dot{V}_z = F_z/m - g$, $\dot{q} = M_{y,\text{CG}}/I_{yy}$ (plus kinematics), and inverse-kinematics-resolves the shoulder/elbow angles so the airframe lands at $(x_{cg}, z_{cg})$. The disk thrust enters separately because it is a volume body force. The moment is translated from a fixed world-frame moment centre to the current CG via $M_{y,\text{CG}} = M_{y,\text{mc}} + (m_z - z_{cg})F_x - (m_x - x_{cg})F_z$, with the moment centre placed at the

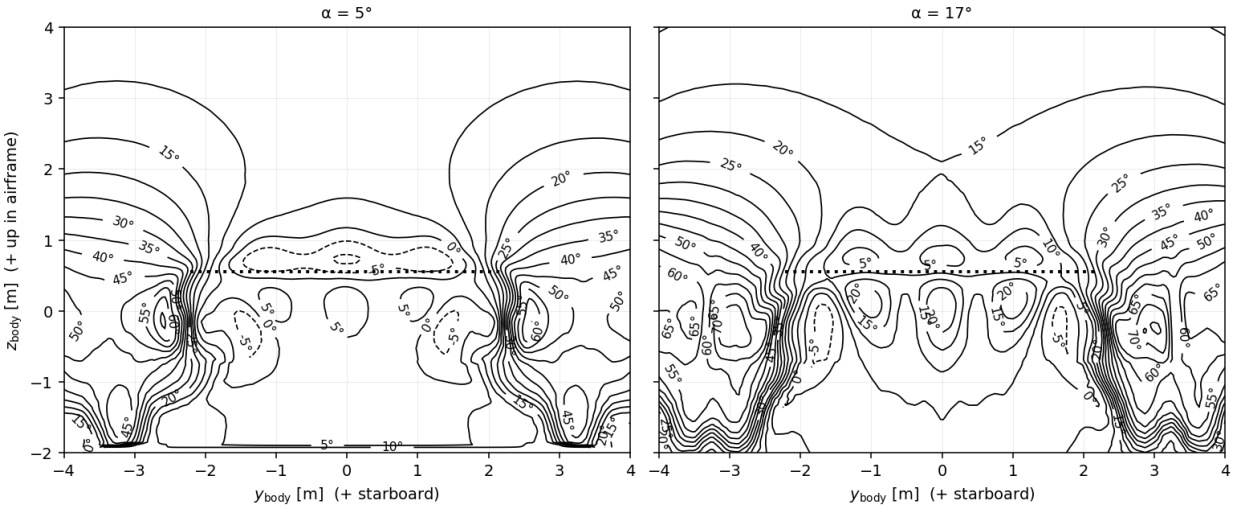



**Fig. 12 Wind-frame downwash $\epsilon$ at the v3 htail station ($x_b$ = htail LE $-0.23$ m, v3 BO landing trim), $\alpha = +5°$ and $+17°$; conventions as Fig. 4. The shortened boom retains the gap-upwash column and the slipstream acceleration at the htail station; combined with the $0.53\times$ moment arm this gives the v3 trim headroom of Fig. 11.**

initial-CG world position to minimise lever-arm cancellation.

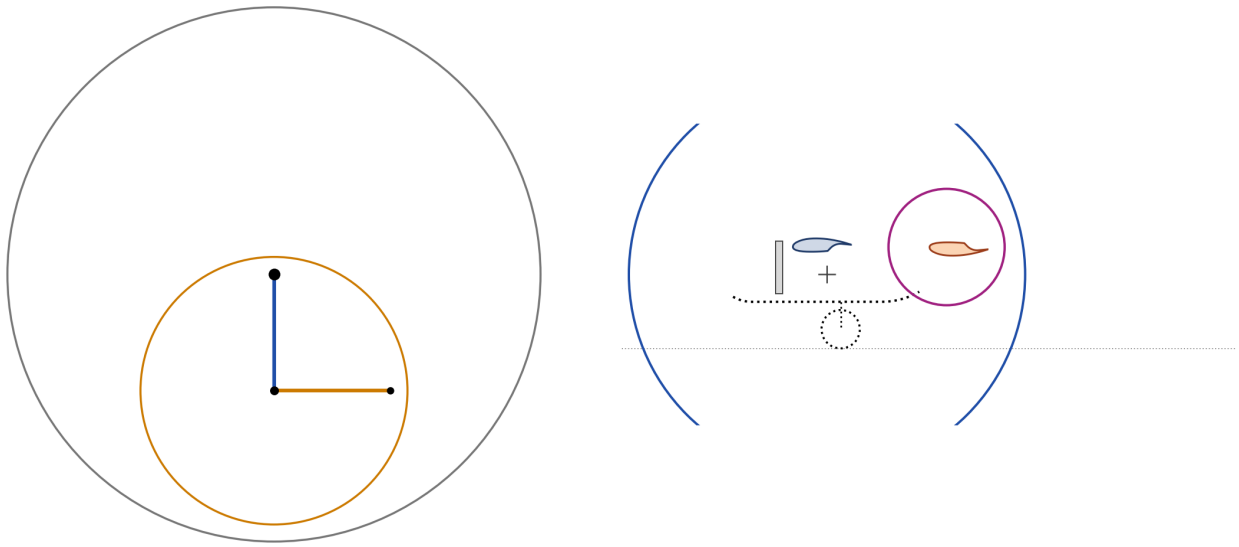


**Fig. 13 Four-cylinder robot-arm topology. Left: world-frame workspace overview; freestream is along $+x$ from the left edge. Cyl 1 (shoulder, grey) and cyl 2 (elbow, orange) form the translating 2-link arm ($L_1 = L_2 = 750\,c_w$); their links are drawn as blue and orange line segments; black dots mark joints. Right: airframe-scale side view in body frame (origin at CG, $+x$ toward tail, $+z$ up). Solid shapes are the geometry meshed in CFD: wing at $x \in [-0.5, +0.5]\,c_w$, htail at $[+1.5, +2.5]\,c_w$, actuator disk at $x = -0.7\,c_w$. Dotted contours (fuselage belly, landing gear, ground) are schematic context not present in the simulation. Cyl 3 (airframe-pitch, blue) is a circle of radius $2.9\,c_w$ centred at the CG – visible as two near-vertical arcs at $x \approx \pm 2.9\,c_w$. Cyl 4 (htail-pitch, magenta) is a circle of radius $0.85\,c_w$ centred at the htail $c/4$ ($x = +1.75\,c_w$), giving a $0.10\,c_w$ design gap between htail TE and cyl 4 on the $+x$ side. Cyl 2 (radius $L_2 + 2.9 + 110 = 863\,c_w$) and cyl 1 (radius $1723\,c_w$) cannot be drawn at this scale: the $110\,c_w$ sliding-interface gaps, set by the mesh-resolution matching rule in `unsteady_setup.py`, put cyl 2's nearest boundary $113\,c_w$ from the CG on the nose side and cyl 1's about $660\,c_w$ away; see the left panel.**

**Table 2 Trimmable flight envelope, v3 short-boom. At each $(\alpha, T/L)$ grid point, $\theta_{ht}$ trims the moment; equilibrium $V$ and $\gamma$ follow from the Newtonian force balance with delivered actuator- disk thrust held constant. $T/L$ is the delivered thrust-to-total-lift ratio at the trimmed airspeed. Highest-$T_{\text{mult}}$ entries at takeoff $\alpha = +12^\circ$ and $+15^\circ$ are omitted: there the drag balance admits no real positive $V$ root (delivered thrust exceeds the largest steady-flight drag at any airspeed), i.e. those points are not steady-flight operating conditions. Slopes from `post/out/v3/*_sweep_data.csv`; table generated by `post/v3/equilibrium_grid.py`.**

| Phase | $\alpha$ [deg] | $T/L$ | $\theta_{ht}$ [deg] | $C_L$ | $V$ [kt] | $\gamma$ [deg] |
|---|---|---|---|---|---|---|
| Cruise | +5 | 0.05 | +5.04 | 0.79 | 92.3 | −1.8 |
| | +5 | 0.09 | +5.16 | 0.80 | 91.5 | +0.5 |
| | +5 | 0.14 | +5.28 | 0.81 | 90.6 | +2.9 |
| | +7 | 0.05 | +4.33 | 0.96 | 83.3 | −2.0 |
| | +7 | 0.09 | +4.45 | 0.97 | 82.6 | +0.5 |
| | +7 | 0.14 | +4.57 | 0.98 | 81.9 | +2.9 |
| | +9 | 0.05 | +3.62 | 1.14 | 76.5 | −2.1 |
| | +9 | 0.09 | +3.74 | 1.15 | 75.9 | +0.4 |
| | +9 | 0.14 | +3.86 | 1.16 | 75.2 | +2.9 |
| Takeoff | +9 | 0.37 | −3.64 | 3.42 | 42.9 | +5.9 |
| | +9 | 0.75 | −3.37 | 3.93 | 37.2 | +22.9 |
| | +9 | 2.54 | −3.10 | 4.44 | 22.6 | +60.1 |
| | +12 | 0.37 | −4.71 | 3.88 | 39.9 | +4.1 |
| | +12 | 0.76 | −4.44 | 4.39 | 34.5 | +22.2 |
| | +15 | 0.37 | −5.78 | 4.34 | 37.3 | +2.8 |
| | +15 | 0.78 | −5.51 | 4.86 | 32.2 | +21.7 |
| Landing | +15 | 0.22 | −2.70 | 4.97 | 35.3 | −10.9 |
| | +15 | 0.40 | −3.18 | 5.83 | 31.9 | −7.1 |
| | +15 | 0.58 | −3.66 | 6.70 | 29.1 | −0.8 |
| | +20 | 0.23 | −3.94 | 5.36 | 33.4 | −13.5 |
| | +20 | 0.40 | −4.42 | 6.23 | 30.2 | −8.4 |
| | +20 | 0.58 | −4.90 | 7.10 | 27.4 | −1.0 |
| | +25 | 0.23 | −5.18 | 5.76 | 31.7 | −15.7 |
| | +25 | 0.40 | −5.66 | 6.63 | 28.6 | −9.6 |
| | +25 | 0.60 | −6.13 | 7.50 | 25.8 | −1.2 |

**Two implementation lessons.** (i) The UDD state update must be gated at the *first* pseudo-step of each physical step (`pseudoStep == 0`), not the last. Flow360 snapshots each zone's previous angle at pseudo-step 0 and rotates the grid by the difference; a last-step gate makes that snapshot copy the already-advanced angle, so the mesh never moves and the wing's lift response to body attitude silently vanishes. At pseudo-step 0 the force registry still holds the previous step's converged values, so the integration is correct with a one-step lag. That lag must be honoured in the moment translation: the lever arm from the fixed moment centre to the CG has to be evaluated at the pose where the forces were computed (forward kinematics of the previous step's link angles), not at the freshly integrated position; using the latter biased $M_{y,\text{CG}}$ enough to flip the sign of the no-kick response. (ii) Pseudo steps per physical step must be large enough for several decades of residual drop; we use 200 (vs. Flow360 default 20–50), reaching a $\sim 0.05\%$ noise floor on $C_{m,\text{mc}}$ – sufficient to expose $\sim 10^3$ N·m $M_{y,\text{CG}}$ signals out of $\sim 10^5$ N·m raw moments.

## V. Expected Full-Paper Contributions

The full SciTech 2027 manuscript (1 December 2026 deadline) will extend this extended abstract with:

- Completion of the v3 and gap-high CFD sweeps (full 1 + 19 cases per phase), sharpening Table 1 and Table 2 and

isolating gap-alone vs. $z$-alone contributions through the $2 \times 2$ matrix.

- Common-$(T/L, \theta_{ht})$-grid rerun across all four v2 configurations to remove the cross-configuration bias of Sec. II, and a body-fixed downwash slice with explicit frame convention.
- Nonlinear phugoid dynamic simulations for cont-high, gap-low v2, and v3 at the landing trim – damping ratio, natural frequency, and step-response time histories, converting the static-margin headline into handling qualities.
- Ground-effect landing flare in CFD: $-3°$ approach, flare at $h = 2$ ft AGL, throttle chop, pitch-up to $+17°$; resolves ground-effect lift amplification, real-time htail authority, and slipstream–ground interaction.
- Asymmetric prop-out sweep + dead-man's-curve characterisation: single- and two-disk-out at the landing trim, centerline vs. outboard cases separated, producing the $(V, h_{\mathrm{AGL}})$ exclusion region. Crucially, each inner-prop-out case will be evaluated for both *stability margin* (SM refit on the de-energised local flow at the htail) and *htail stall margin* (recomputed effective $\alpha_{ht}$ now that the centerline upwash column is partially collapsed) – both can degrade substantially even when the airframe remains trimmable.
- Actuator-disk validation: a swirl-augmented disk variant and a rotor-resolving sliding-mesh spot-check at v3 landing [9, 10], bounding swirl/helical-tip-vortex error on the symmetric upwash column.
- Acoustic spot-check at design tip-Mach and a gap-fraction sensitivity (fixed at 0.40 here).
- Numerical practice for the four-cylinder dynamic-CFD framework: meshing strategy across the nested rotation zones, residual-convergence behaviour as a function of pseudo-steps per physical step, wall-time cost per simulated second on the phugoid and the landing-flare cases, and the sensitivity of the integrated CG moment to the moment-translation cancellation discussed above.

**Mapping the stability paradox across design space.** The $2 \times 2$ design matrix of §II is a four-point sample of a continuous five-dimensional design space $(f_{\mathrm{gap}}, z_{ht}, x_{ht}, \alpha, T/L)$. A central question this abstract cannot yet answer is whether the gap-low stability paradox is a finite-volume basin robust to perturbation along each axis, or a knife-edge that the $2 \times 2$ happens to straddle. The full paper will recover the boundary of the paradox region (where the landing SM crosses zero) via an adaptive-sampling campaign: an AVL vortex-lattice linear prior, a residual Gaussian-process correction fit to acquired Flow360 cases, and an acquisition layer that preferentially queries near sign reversals in $\partial C_{m_{\mathrm{CG}}}/\partial\alpha$, supplemented by a compact learned next-point policy trained on synthetic aerodynamic databases of known ground truth. The architecture has been validated on a 2,333-case BET-RANS oracle of the NASA X-57 Maxwell, recovering a previously unresolved nonlinear stall transition at $\sim$ 30–45% value-RMSE improvement over Latin-hypercube sampling at matched budget, with $|\nabla C_L|$ RMSE $\sim$ 52% lower than LHS at small budgets.* Reconstruction quality will be reported in a Sobolev-type norm $\|\hat{C}_{m_{\mathrm{CG}}} - C^{\star}_{m_{\mathrm{CG}}}\|_{H^1}$ on the slope and curvature of $C_{m_{\mathrm{CG}}}$, which is the engineering-relevant metric for a stability-margin question and where the adaptive sampler's advantage is largest. The campaign is orchestrated end-to-end in Flexcompute Thread,† which also exposes derived engineering quantities (static margin, sensitivity slopes, Sobolev-norm reconstruction error) alongside the primitive CFD coefficients. Deliverables: (i) the recovered paradox-region boundary surface in the 5-D space; (ii) Sobolev-norm convergence of reconstructed $C_{m_{\mathrm{CG}}}$ vs adaptive case count $N_{\mathrm{adaptive}}$, with the equivalent uniform-grid budget $N_{\mathrm{uniform}}$ as the comparison anchor; and (iii) a held-out blind verification point at which the learned policy predicted the boundary location, used to demonstrate generalisation outside the training set.

## Reproducibility and use of AI

All Flow360 case identifiers, geometry CSM/UDC sources, and post-processing scripts are committed to the public project repository `github.com/qiqi/tsangpo`; this document was built from commit `e5d07cc-dirty`. The authors used Anthropic's Claude as an iterative engineering assistant for prose drafting, Python scripting (case setup, mesh configuration, post-processing, animation), Flow360 UDD authoring ($\theta/\omega/\dot{\omega}$ publishing in non-dimensional solver units), and co-design of the four-cylinder nested- rotation mesh topology and its inverse-kinematics formulation for the unsteady cases. All technical decisions, interpretation of results, and final responsibility for the content rest with the human authors, in compliance with AIAA's policy on AI in scholarly publications.

*Flexcompute internal study; details to appear in the full paper.

†Flexcompute Thread, `https://thread.simulation.cloud`: integrated data-management framework that unifies the AVL prior, Flow360 RANS, the residual GP, and the learned next-point policy as queryable objects of the same design study.

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## A. Geometry Reference

All units SI (m, kg, N, s, rad); body frame at CG, $+x$ aft, $+y$ starboard, $+z$ up. The wing is a Hershey-bar planform (rectangular, no taper, sweep, twist, or dihedral) with the three-element high-lift section of Fig. 1 at every spanwise station[‡]. The vane and aft flap deploy as a rigid assembly hinged at $(0.550, -0.048)\, c_w$, with phase translations/rotations $(\Delta x, \Delta y, \Delta\theta)$ of $(0, 0, 0°)$ stowed, $(0.280, 0.045, -40°)$ takeoff, $(0.300, 0.050, -65°)$ landing ($\Delta\theta$ TE-down positive here). The flap-gap variants delete the vane+aft-flap assembly on $|y| < 0.20\, b$[§]. The horizontal tail is rectangular, inverted LS(1)-0417 section (negative camber), pivoted about its $c/4$. Each propeller disk is a constant-force actuator disk normal to body $x$ with a $0.1\, c_w$ axial mesh region; $T_{\text{mult}}$ scales every disk's pressure jump uniformly and results are reported as the resulting delivered $T/L = C_T/C_{L,\text{total}}$.

[‡]Main element: modified LS(1)-0417 with a coved lower-surface cutout starting at $x/c = 0.510$, terminating at the cove vertex $(0.620, 0.055)$ with a $0.2\, c$-radius fillet, and an upper-surface lip cut at $x/c = 0.850$. Vane: NACA 9621, stowed LE/TE $(0.550, -0.048)/(0.730, +0.035)\, c_w$, element chord $\approx 0.198\, c_w$. Aft flap: NACA 6311, stowed LE/TE $(0.690, -0.011)/(1.000, 0.000)\, c_w$, element chord $\approx 0.310\, c_w$. Uniform $0.003\, c_w$ blunt-TE thickness on all three elements.

[§]Main element continuous across the gap; inboard and outboard flap segments retain their Fowler kinematics, separated from the no-flap region by a $0.005\, b$ spanwise air gap for mesh-discrete edge separation in CFD.

**Table 3 Geometry parameters (lengths in m; symbol/value).**

| Quantity | Symbol | Value |
|---|---|---|
| *Wing and weight* | | |
| Reference area | $S_w$ | 15.33 ($\approx 165\,\mathrm{ft}^2$) |
| Span | $b$ | 11.074 |
| Aspect ratio | AR | 8.0 |
| MAC | $c_w$ | 1.385 |
| Wing LE $x$ | $x_{LE,w}$ | $-0.692$ ($= -0.5\,c_w$) |
| Wing $c/4$ $x$ | $x_{c/4,w}$ | $-0.346$ |
| Wing chord plane $z$ | $z_w$ | $+0.554$ ($= +0.4\,c_w$ above CG) |
| Gross weight | $W$ | 11,565 N ($\approx$ 2,600 lbf) |
| *Flap gap (gap variants)* | | |
| Gap half-width | – | $0.20\,b = 2.215$ |
| *Horizontal tail* | | |
| Span | $b_{ht}$ | 4.430 ($= 0.40\,b$) |
| Chord | $c_{ht}$ | 1.385 ($= 1.00\,c_w$) |
| Area | $S_{ht}$ | 6.135 ($S_{ht}/S_w = 0.40$) |
| LE $x$ | $x_{LE,ht}$ | $+4.847$ ($= +3.5\,c_w$, $4\,c_w$ aft of wing LE) |
| $c/4$ $x$ | $x_{c/4,ht}$ | $+5.193$ ($= +3.75\,c_w$ for v2; $+2.0\,c_w$ for v3) |
| $z$ low / high | $z_{ht}$ | $+0.554$ / $+2.631$ |
| Tail volume | $V_H$ | $\approx 1.60$ |
| *Propeller array: 10 disks, mirror-symmetric in $y$, at $\eta = y/(b/2) \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$* | | |
| Disk diameter | $D$ | 1.067 |
| Disk radius | $R$ | 0.534 ($= 0.385\,c_w$) |
| Disk $x$ | $x_{prop}$ | $-0.969$ ($= -0.7\,c_w$) |
| Disk $z$ | $z_{prop}$ | $+0.138$ ($= +0.1\,c_w$) |
| Disk $y$ | $y_{prop}$ | $\pm\{0.554, 1.661, 2.769, 3.876, 4.983\}$ |
| Design $T/W$ | – | 0.5 |
| Thrust per disk @ design | $T_p$ | 578 N |
| Disk loading @ design | $DL$ | 1,292 N/m$^2$ |